

PROJECT TECHNICAL REPORT

SkyCast

DESIGN AND IMPLEMENTATION OF A MACHINE LEARNING-BASED MULTI-HORIZON
AIR QUALITY FORECASTING SYSTEM FOR URBAN ENVIRONMENTS: A CASE STUDY OF
KARACHI

Air Quality Index (AQI) Forecasting Platform

Karachi, Pakistan

Table of Contents

PART 1: ABSTRACT & EXECUTIVE SUMMARY	4
1.1 Executive Overview	4
1.2 Data Acquisition, Extraction, and Stream Integration	4
1.2.1 Multimodal Telemetry Profiles	5
Atmospheric Composition Profiles	5
Meteorological Boundary Conditions	6
1.2.2 Temporal Joining and Consolidation	6
Unified Timestamp Alignment	6
Inner Merge Integration	6
1.3 System Innovation and Operational Value	6
PART 2: REPOSITORY STRUCTURE & TECHNICAL ARCHITECTURE	7
2.1 System Repository Topography	7
Repository Structure	8
Directory and Module Description	8
2.2 Architectural Significance	11
2.2 Data Acquisition, Extraction, and Ingestion Engine	11
Air Quality Data Extraction	12
Weather Telemetry Extraction	12
Deterministic Temporal Alignment	12
Unified Data Consolidation	13
2.3 Comprehensive Platform Dependency Configuration	13
Core Platform Dependencies	13
Dependency Architecture Significance	14
PART 3: METHODOLOGY, FEATURE ENGINEERING & MODELING RESULTS	15
3.1 Predictive Modeling Strategy and Target Formulation	15
3.2 Advanced Feature Engineering Store	15
Feature Categories	16
3.3 Multi-Horizon Model Training and Evaluation	19
Mean Absolute Error (MAE)	19
Root Mean Squared Error (RMSE)	19
Coefficient of Determination (R^2)	19
Model Performance Summary	19
3.3.1 Model Selection and Production Deployment	20
3.4 Feature Importance Analysis of the Production Model	20
Top Influential Features	20
3.4.1 Insights from Feature Rankings	21
PART 4: EXPLAINABLE AI (XAI) THROUGH SHAP INTERPRETATION	22
4.1 Foundations of Model Interpretability	22
4.2 SHAP Summary Global Interpretation (24-Hour Horizon)	22
Primary Particulate Matter Indicators	22
Pollution Volatility Indicators	23
Seasonal and Climatic Effects	23
4.3 Middle-Horizon Shift Analysis (48-Hour Horizon)	23
Persistent Air Quality Trends	23
Role of Secondary Pollutants	23
Baseline Pollution Constraints	23
4.4 Long-Horizon Stability Analysis (72-Hour Horizon)	24

Atmospheric Persistence and Long-Term Trends.....	24
Meteorological Stability Factors.....	24
Wind and Pressure Dynamics.....	24
4.5 Multi-Horizon Interpretability Summary.....	24
PART 5: LOCAL INTERPRETABILITY VIA LIME EXPLANATIONS	25
5.1 Objectives of Localized Instance Explanations	25
5.2 Decoupling the Dashboard Visual Matrix.....	25
Interpretation of LIME Contribution Bars	25
5.3 Analysis of Primary Risk Ingestion (Case Study 1)	26
Sustained Baseline Pollution	26
Chemical Saturation Effects	26
Secondary Pollutant Contributions	26
5.4 Analysis of Atmospheric Mitigation (Case Study 2).....	26
Seasonal and Cyclical Influence.....	26
Reduction in Particulate Concentration	26
Meteorological Dispersion Effects.....	27
5.5 Combined Interpretability Framework.....	27
PART 6: PRODUCTION USER INTERFACE & COMPREHENSIVE DASHBOARD ECOSYSTEM	27
6.1 Architectural Overview of the User Interface	27
6.2 Real-Time Hazard Warnings and Categorization Mechanics.....	28
6.3 Interactive Index Charting Matrix.....	29
Key Features of the Visualization System.....	29
6.4 Analytical Intelligence and Export Pipelines.....	29
System Functions.....	29
6.4.1 Automated Executive Report Generator ("Generate AI Forecast")	30
6.4.2 Flattened Time-Series Data Export ("Save as CSV")	30
Inverse Transformation Mechanism	30
6.5 Final System Summary	30

PART 1: ABSTRACT & EXECUTIVE SUMMARY

1.1 Executive Overview

Urban air pollution has become one of the most critical public health and environmental challenges faced by modern metropolitan cities. Poor air quality significantly impacts human health, increases healthcare costs, and affects overall quality of life. To address this challenge, **SkyCast** has been developed as a cloud-native, machine learning-powered Air Quality Index (AQI) forecasting platform for Karachi, Pakistan.

Unlike traditional air quality monitoring systems that only provide current or historical pollution measurements, SkyCast delivers proactive environmental intelligence by forecasting AQI levels for the next **24, 48, and 72 hours**. The platform combines advanced machine learning algorithms, automated MLOps pipelines, and an interactive web-based dashboard to transform large volumes of environmental data into actionable insights for citizens, researchers, and decision-makers.

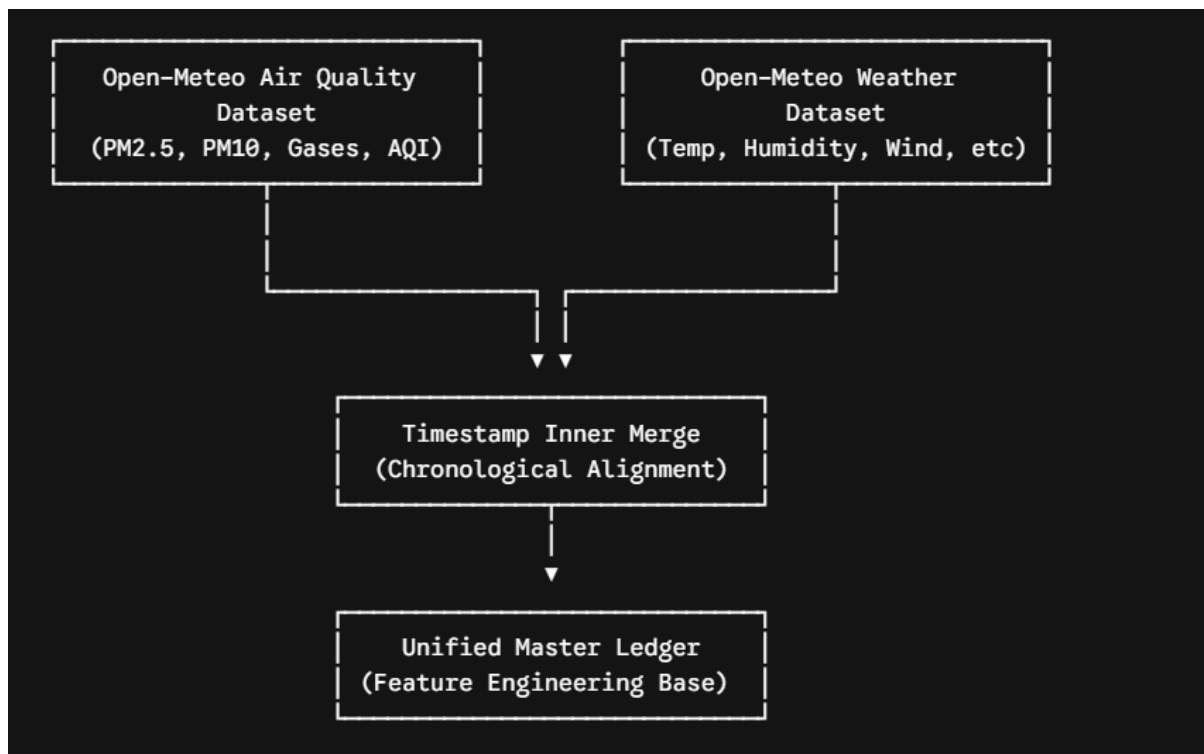
By providing early warnings of deteriorating air quality conditions, SkyCast enables individuals and organizations to take preventive measures, thereby supporting public health protection and environmental awareness. The system demonstrates how modern Artificial Intelligence and cloud technologies can be leveraged to solve real-world environmental challenges through accurate and scalable predictive analytics.

1.2 Data Acquisition, Extraction, and Stream Integration

The predictive capability of SkyCast is built upon a comprehensive environmental dataset collected from the **Open-Meteo Air Quality API** and **Open-Meteo Historical Weather API**. The dataset spans approximately **3.25 years of continuous historical records**, providing a rich foundation for identifying seasonal trends, atmospheric patterns, and pollution behavior.

To ensure data consistency and completeness, a dedicated data engineering pipeline was developed to automate the collection, validation, and integration of multiple environmental data streams. The pipeline retrieves both air quality indicators and weather-related variables before merging them into a unified analytical dataset.

The integration process follows the workflow illustrated below:



The resulting master dataset serves as the central repository for all downstream machine learning and forecasting operations. By combining atmospheric pollutant measurements with meteorological conditions, the platform captures the complex relationships that influence air quality dynamics and enhances the predictive performance of the forecasting models.

1.2.1 Multimodal Telemetry Profiles

To accurately model air quality dynamics, the SkyCast platform collects and integrates environmental information from two complementary telemetry streams: atmospheric composition data and meteorological observations. This multimodal approach enables the system to capture both pollution levels and the environmental factors that influence their behavior.

Atmospheric Composition Profiles

The atmospheric composition stream monitors the concentration of major air pollutants that directly contribute to air quality degradation and public health risks. These pollutants include:

- Fine Particulate Matter (PM2.5)
- Coarse Particulate Matter (PM10)
- Nitrogen Dioxide (NO₂)
- Carbon Monoxide (CO)
- Sulphur Dioxide (SO₂)
- Ground-Level Ozone (O₃)

In addition to pollutant concentrations, the dataset includes calculated United States Air Quality Index (US AQI) benchmark values, providing standardized indicators of air quality severity and associated health impacts.

Meteorological Boundary Conditions

Air pollution behavior is strongly influenced by prevailing weather conditions. Therefore, the second telemetry stream captures critical meteorological variables, including:

- Surface Temperature
- Relative Humidity
- Wind Speed
- Wind Direction
- Cloud Cover
- Precipitation
- Sea-Level Atmospheric Pressure

These environmental factors play a significant role in pollutant transport, dispersion, accumulation, and chemical transformation within the atmosphere.

1.2.2 Temporal Joining and Consolidation

Because air quality conditions are closely linked to weather patterns, both telemetry streams were integrated into a single analytical dataset using a deterministic time-series alignment process.

Unified Timestamp Alignment

The data ingestion pipeline standardizes all incoming records onto a consistent hourly temporal framework. This process ensures synchronization between air quality and weather observations while eliminating inconsistencies caused by differences in collection intervals across the 3.25-year historical period.

Inner Merge Integration

After temporal standardization, the datasets are combined using an exact timestamp-based inner join operation:

```
pd.merge(on="timestamp", how="inner")
```

This merging strategy ensures that only records with matching timestamps are retained, resulting in a complete and highly reliable environmental dataset. The integrated data repository establishes direct relationships between meteorological conditions and pollutant concentrations, allowing the machine learning models to learn how weather variations affect air quality patterns across Karachi.

The resulting unified dataset serves as the primary source for feature engineering, model training, forecasting, and analytical reporting.

1.3 System Innovation and Operational Value

One of the key strengths of SkyCast is its ability to learn from more than three years of historical environmental observations. This extensive dataset enables the system to identify and model complex atmospheric behaviors, including seasonal pollution cycles, coastal wind circulation

patterns, temperature-driven inversions, and other environmental characteristics unique to Karachi's coastal climate.

The platform connects this consolidated environmental data repository to a fully automated machine learning pipeline that continuously updates forecasting models through scheduled cloud-based workflows. This ensures that predictions remain relevant, accurate, and responsive to evolving environmental conditions.

To make advanced predictive analytics accessible to non-technical users, SkyCast provides a range of user-centric features, including:

- Automated AI-generated forecast summaries
- CSV export functionality for data auditing and analysis
- Real-time US AQI category classification
- Health risk notifications and hazard alerts
- Interactive forecasting dashboard for visualization and decision support

These capabilities transform complex environmental datasets into actionable intelligence for citizens, healthcare organizations, researchers, environmental agencies, and policy-makers. By providing reliable short-term AQI forecasts and health advisories, SkyCast supports proactive decision-making and contributes to improved public health awareness and environmental management.

PART 2: REPOSITORY STRUCTURE & TECHNICAL ARCHITECTURE

2.1 System Repository Topography

To ensure scalability, maintainability, and efficient deployment, the SkyCast platform follows a modular and production-oriented repository structure. The codebase has been designed to support automated data ingestion, feature engineering, model training, cloud synchronization, model versioning, and web-based forecasting services within a unified development environment.

The repository organization separates data processing, machine learning workflows, dashboard development, and MLOps operations into dedicated modules. This architectural approach simplifies maintenance, enhances code reusability, and enables independent updates to system components without affecting overall platform functionality.

Repository Structure

```
KARACHI-AQI-FORECASTING-PLATFORM/
├── .github/
│   └── workflows/
│       ├── daily_training_pipeline.yml
│       └── hourly_pipeline.yml
├── dashboards/
│   └── streamlit_app.py
├── data/
│   ├── processed/
│   │   └── aqi_feature_store.csv
│   └── raw/
│       └── karachi_aqi_historical.csv
├── models/
├── notebooks/
├── reports/
├── src/
│   ├── model_registry/
│   │   └── register_model.py
│   ├── data_ingestion.py
│   ├── feature_engineering.py
│   ├── predict.py
│   ├── retrain_pipeline.py
│   ├── train_model.py
│   └── upload_to_supabase.py
├── utils.py
├── .cache_sqlite
├── .env
├── .gitignore
├── LICENSE
├── README.md
└── requirements.txt
```

Directory and Module Description

.github/workflows/

This directory contains the automated GitHub Actions workflows responsible for orchestrating the platform's MLOps lifecycle.

- **daily_training_pipeline.yml** automates daily model retraining, evaluation, and deployment activities.
- **hourly_pipeline.yml** executes scheduled data collection tasks to continuously update environmental observations from external APIs.

These workflows eliminate manual intervention and ensure that both datasets and machine learning models remain up-to-date.

dashboards/

The dashboard layer contains the primary user-facing application.

- **streamlit_app.py** implements the interactive Streamlit dashboard used for AQI forecasting visualization, health alerts, trend analysis, AI-generated summaries, and forecast reporting.

This component serves as the interface between the predictive engine and end users.

data/

The data directory stores both raw and processed datasets used throughout the machine learning lifecycle.

Raw Data Layer

- **karachi_aqi_historical.csv** contains the merged historical environmental dataset collected from Open-Meteo APIs.

This file serves as the master telemetry repository before transformation and feature engineering.

Processed Data Layer

- **aqi_feature_store.csv** contains engineered features ready for machine learning model consumption.

The processed dataset includes lag variables, moving averages, cyclical temporal encodings, and meteorological predictors.

models/

This directory stores serialized machine learning artifacts and trained model weights generated during training and retraining operations.

Maintaining model binaries separately enables rapid deployment, rollback capabilities, and version-controlled inference.

notebooks/

The notebooks directory contains experimental and exploratory development notebooks used during the research and prototyping phase of the project.

Typical activities include:

- Exploratory Data Analysis (EDA)
- Feature importance investigations
- Model comparison experiments
- Data quality validation

reports/

This directory stores generated analytical reports, evaluation summaries, performance benchmarks, and model diagnostics.

These reports provide evidence for model validation and support ongoing monitoring of forecasting performance.

src/

The source directory contains the core business logic and machine learning implementation modules.

model_registry/

This submodule manages model versioning and experiment tracking.

- **register_model.py** integrates with DagsHub and MLflow to automatically register trained models, store metadata, and maintain reproducible model histories.

This functionality ensures proper governance of machine learning assets throughout the deployment lifecycle.

data_ingestion.py

Responsible for collecting and updating historical and real-time environmental data from Open-Meteo APIs. The module performs API communication, validation, cleaning, and dataset generation.

feature_engineering.py

Generates advanced predictive variables used by the machine learning models, including:

- Temporal lag features
- Rolling statistical windows
- Moving averages
- Seasonal indicators
- Cyclical date-time encodings
- Weather interaction features

These engineered variables significantly improve forecasting accuracy.

predict.py

Implements the production inference engine responsible for generating AQI forecasts across multiple prediction horizons.

The module loads the latest registered model and produces operational forecasts displayed within the dashboard.

retrain_pipeline.py

Acts as the orchestration layer for automated model retraining. It coordinates data preparation, feature generation, model training, evaluation, and deployment into a single reproducible workflow.

train_model.py

Contains the primary machine learning training logic, including:

- Hyperparameter optimization
- Cross-validation procedures
- Model fitting
- Performance evaluation
- Metric generation

This module serves as the computational core of the forecasting engine.

upload_to_supabase.py

Synchronizes newly collected environmental observations and prediction outputs to the cloud database hosted on Supabase.

This ensures centralized storage and accessibility across deployment environments.

Utility and Configuration Files**utils.py**

Contains reusable helper functions shared across multiple system modules, reducing code duplication and improving maintainability.

.cache_sqlite

Provides a lightweight caching layer that improves application responsiveness by reducing repetitive database and API requests.

.env

Stores sensitive configuration variables and cloud service credentials, ensuring secure access to external infrastructure components.

.gitignore

Defines repository exclusion rules to prevent temporary files, credentials, and large generated artifacts from being committed to version control.

README.md

Provides project documentation, installation procedures, system architecture explanations, and deployment instructions.

requirements.txt

Lists all Python package dependencies required to reproduce the project environment and execute the platform successfully.

2.2 Architectural Significance

The repository structure reflects a production-grade machine learning system rather than a traditional academic prototype. By separating data engineering, model development, deployment workflows, cloud synchronization, and user interface components into dedicated modules, the platform achieves high maintainability, scalability, and reproducibility.

This modular architecture enables automated operation through MLOps practices while ensuring that each subsystem can evolve independently. As a result, SkyCast remains capable of supporting continuous environmental monitoring, daily model updates, cloud deployment, and real-time AQI forecasting within a unified and scalable framework.

2.2 Data Acquisition, Extraction, and Ingestion Engine

The analytical foundation of the SkyCast platform is built upon approximately 3.25 years of continuous historical environmental observations collected from the Open-Meteo Air Quality and

Historical Weather databases. To automate the collection and consolidation of this information, the system utilizes a dedicated data ingestion module implemented in **src/data_ingestion.py**.

This module functions as an automated data harvesting engine that retrieves, validates, and integrates multiple environmental data streams into a unified analytical dataset suitable for machine learning operations.

Air Quality Data Extraction

The ingestion engine initiates programmatic API requests using Karachi's geographical coordinates:

- Latitude: 24.8607
- Longitude: 67.0011

Using these coordinates, the system retrieves historical measurements for key atmospheric pollutants, including:

- Fine Particulate Matter (PM2.5)
- Coarse Particulate Matter (PM10)
- Carbon Monoxide (CO)
- Nitrogen Dioxide (NO₂)
- Ground-Level Ozone (O₃)
- Sulphur Dioxide (SO₂)

Additionally, the platform acquires corresponding air quality benchmark indicators used for regulatory assessment and AQI calculation.

Weather Telemetry Extraction

To capture the environmental factors that influence pollutant behavior, the ingestion engine simultaneously retrieves historical meteorological observations. The extracted variables include:

- Ambient Air Temperature
- Relative Humidity
- Wind Speed
- Atmospheric Pressure

These weather parameters provide essential contextual information that allows forecasting models to learn the relationship between atmospheric conditions and air pollution levels.

Deterministic Temporal Alignment

Because pollutant concentrations and weather conditions evolve simultaneously over time, accurate synchronization between both datasets is critical.

The ingestion framework standardizes all incoming observations onto a consistent hourly timeline. This process ensures temporal consistency across the entire historical observation period and eliminates discrepancies caused by varying data collection frequencies.

Unified Data Consolidation

Following temporal alignment, both datasets are combined using an exact timestamp-based inner merge operation:

```
pd.merge(aq_df, weather_df, on="timestamp", how="inner")
```

This deterministic merging strategy guarantees that only records with perfectly matching timestamps are retained. As a result, the platform eliminates missing temporal relationships and ensures high-quality analytical consistency throughout the historical dataset.

The final merged dataset is stored as:

```
data/raw/karachi_aqi_historical.csv
```

This file serves as the master environmental data repository and forms the basis for feature engineering, model training, forecasting, and performance evaluation activities.

2.3 Comprehensive Platform Dependency Configuration

To support large-scale data processing, machine learning operations, cloud integration, and interactive visualization, the SkyCast platform relies on a carefully selected set of software libraries and frameworks. Each dependency performs a specialized role within the overall system architecture.

Core Platform Dependencies

Package Dependency	Production Purpose and Functional Contribution
◇ streamlit ($\geq 1.32.0$)	Provides the interactive dashboard environment used for AQI visualization, forecasting outputs, trend analysis, and user interaction.
◇ supabase	Enables secure cloud database communication for storing and retrieving environmental observations, engineered features, and forecast outputs.
◇ openmeteo-requests	Facilitates communication with Open-Meteo APIs and automates retrieval of historical and real-time environmental data.

◇ scikit-learn	Supplies model evaluation metrics, preprocessing utilities, validation frameworks, and performance measurement tools.
◇ pandas	Supports structured data manipulation, transformation, aggregation, and feature generation across large time-series datasets.
◇ numpy	Performs high-speed numerical computation and vectorized mathematical operations required for machine learning workflows.
◇ joblib	Serializes trained machine learning models into compact binary files for efficient storage and rapid deployment.
◇ dagshub	Provides cloud-based experiment tracking, collaboration features, and machine learning project management capabilities.
◇ mlflow	Maintains model lineage, version control, experiment tracking, and centralized model registry operations.

Dependency Architecture Significance

The selected technology stack was designed to support the complete machine learning lifecycle, including data acquisition, preprocessing, feature engineering, model development, deployment, monitoring, and visualization.

By combining cloud-native services, modern machine learning frameworks, and automated MLOps tooling, the platform achieves:

- High forecasting accuracy
- Automated data synchronization
- Scalable cloud deployment
- Reproducible machine learning workflows
- Reliable model version control
- Efficient dashboard responsiveness

This dependency ecosystem enables SkyCast to function as a fully integrated predictive climate intelligence platform capable of delivering continuous AQI forecasting services with minimal operational overhead.

PART 3: METHODOLOGY, FEATURE ENGINEERING & MODELING RESULTS

3.1 Predictive Modeling Strategy and Target Formulation

The primary objective of SkyCast is to provide proactive air quality intelligence by forecasting future Air Quality Index (AQI) conditions across multiple prediction horizons. To achieve this objective, the system was designed to generate forecasts for three distinct future windows:

- 24-Hour AQI Forecast
- 48-Hour AQI Forecast
- 72-Hour AQI Forecast

From a machine learning perspective, this problem is formulated as a supervised time-series regression task. Rather than directly predicting categorical AQI levels, the forecasting models operate on continuous environmental measurements. This approach preserves numerical precision during training and avoids information loss that may occur when continuous pollution measurements are converted into discrete AQI categories prematurely.

The modeling framework predicts future particulate matter concentrations using historical pollution observations and meteorological conditions. Three independent target variables were created:

- target_24h
- target_48h
- target_72h

These targets represent future atmospheric pollution levels at 24-hour, 48-hour, and 72-hour forecasting horizons respectively.

After generating continuous predictions, the results are converted into standardized United States Air Quality Index (US AQI) values through post-processing functions. These AQI values are then presented to users through the dashboard along with corresponding health advisory categories.

3.2 Advanced Feature Engineering Store

The historical environmental dataset was transformed into a comprehensive analytical feature store through the feature engineering pipeline implemented in **src/feature_engineering.py**.

Feature engineering plays a critical role in time-series forecasting because environmental pollution patterns are influenced by temporal cycles, historical pollutant concentrations, weather conditions, and long-term seasonal trends. To capture these relationships, multiple categories of predictive features were generated.

Feature Categories

Geographic and Timestamp Features

These variables provide the fundamental spatial and temporal reference for each observation.

- city
- timestamp

Atmospheric Pollutant Features

These variables represent measured concentrations of key pollutants.

- pm2_5
- pm10
- carbon_monoxide
- nitrogen_dioxide
- ozone
- sulphur_dioxide

Air Quality Index Variables

- us_aqi
- european_aqi

Meteorological Features

Weather conditions that influence pollutant formation and dispersion.

- temperature
- humidity
- wind_speed
- pressure
- precipitation
- cloudcover

Temporal and Cyclical Encodings

To capture recurring environmental patterns, cyclical representations of time were generated.

- hour_of_day
- day_of_week
- month_of_year
- day_of_month
- week_of_year
- is_weekend
- season
- hour_sin
- hour_cos
- day_sin

- day_cos
- month_sin
- month_cos

Mathematical Transformations

Additional transformed variables were generated to capture nonlinear pollutant behavior.

- pm25_log
- pm_ratio

Historical Lag Features

Lag features allow the model to learn how previous pollution levels influence future conditions.

Examples include:

- aqi_lag_1h
- aqi_lag_6h
- aqi_lag_24h
- pm25_lag_1
- pm25_lag_2
- pm25_lag_3
- pm25_lag_6
- pm25_lag_12
- pm25_lag_24
- pm25_lag_48

Rolling Statistical Windows

Moving window calculations were generated to capture short-term environmental trends.

Examples include:

- aqi_rolling_avg_6h
- aqi_rolling_avg_24h
- aqi_change_rate_24h
- pm25_roll_mean_24
- pm25_roll_std_24
- pm25_roll_max_24
- pm25_roll_min_24
- pm25_median_24

Exponential and Volatility Features

These variables emphasize recent observations while measuring atmospheric variability.

Examples include:

- pm25_ema_3

- pm25_ema_6
- pm25_ema_12
- pm25_ema_24
- pm25_ewm_std_24
- pm25_volatility_24
- pm25_cumulative_24

Pollution Event Indicators

Specialized indicators were introduced to identify unusual pollution behavior.

- pollution_spike
- high_pollution_flag

Weather Lag Features

Historical weather observations were incorporated to capture delayed environmental effects.

Examples include:

- temperature_lag_6
- temperature_roll_mean_24
- humidity_lag_6
- humidity_roll_mean_24
- pressure_lag_6
- pressure_roll_mean_24
- wind_speed_lag_6
- wind_speed_roll_mean_24

Chemical Pollutant Lag Features

Historical concentrations of secondary pollutants were also included.

Examples include:

- carbon_monoxide_lag_6
- carbon_monoxide_roll_mean_24
- nitrogen_dioxide_lag_6
- nitrogen_dioxide_roll_mean_24
- sulphur_dioxide_lag_6
- sulphur_dioxide_roll_mean_24
- ozone_lag_6
- ozone_roll_mean_24

Multi-Horizon Prediction Targets

The final feature store includes three forecasting targets:

- target_24h
- target_48h

- target_72h

This extensive feature engineering process enabled the machine learning models to learn complex relationships between weather conditions, pollutant dynamics, and seasonal environmental trends.

3.3 Multi-Horizon Model Training and Evaluation

To identify the most suitable forecasting architecture, five machine learning algorithms were trained and evaluated using the complete historical dataset.

The evaluated models included:

1. Extra Trees Regressor
2. Random Forest Regressor
3. XGBoost Regressor
4. LightGBM Regressor
5. Long Short-Term Memory (LSTM) Neural Network

Performance was assessed using three standard regression metrics:

Mean Absolute Error (MAE)

Measures the average prediction error magnitude.

Root Mean Squared Error (RMSE)

Penalizes larger prediction errors and provides insight into forecast stability.

Coefficient of Determination (R^2)

Measures how effectively the model explains variation within the target variable.

Model Performance Summary

The evaluation results demonstrated that tree-based ensemble methods consistently outperformed deep learning and gradient boosting alternatives across all forecasting horizons.

Among all evaluated algorithms, the Extra Trees Regressor achieved the strongest overall performance.

Best Performing Model Results (Extra Trees)

Forecast Horizon	MAE	RMSE	R^2
24 Hours	8.658	11.983	0.336
48 Hours	9.475	12.726	0.239
72 Hours	9.694	12.754	0.246

The model maintained stable prediction accuracy even as the forecasting horizon extended from one day to three days.

3.3.1 Model Selection and Production Deployment

Based on the evaluation results, the Extra Trees Regressor was selected as the production forecasting model.

The selection was driven by:

- Lowest Mean Absolute Error across all horizons
- Strong RMSE stability
- Consistent predictive performance
- Robust handling of nonlinear environmental relationships
- Reduced susceptibility to overfitting

After final training, the models were serialized and stored as deployable artifacts.

Successfully generated production models include:

- best_model_24h.pkl
- best_model_48h.pkl
- best_model_72h.pkl

These artifacts serve as the operational forecasting engines used by the Streamlit dashboard and cloud deployment infrastructure.

3.4 Feature Importance Analysis of the Production Model

To improve model interpretability and understand the environmental factors driving AQI predictions, feature importance analysis was performed on the trained Extra Trees model.

Feature importance scores quantify how much each variable contributes to prediction accuracy during tree construction.

The analysis revealed that seasonal indicators, pollution history, atmospheric conditions, and chemical pollutant interactions all play significant roles in determining future air quality levels.

Top Influential Features

The most influential variables identified by the model include:

1. month_cos
2. high_pollution_flag
3. season
4. pm_ratio
5. ozone

6. pm25_log
7. month_sin
8. pm2_5
9. aqi_lag_24h
10. pm25_lag_24

Additional influential variables include rolling averages, volatility metrics, sulphur dioxide levels, carbon monoxide concentrations, nitrogen dioxide concentrations, and long-term particulate matter trends.

3.4.1 Insights from Feature Rankings

The feature importance analysis provides valuable insight into the environmental processes governing air quality in Karachi.

Seasonal Effects as Dominant Drivers

The strong importance of variables such as month_cos, month_sin, and season indicates that air quality is heavily influenced by seasonal environmental cycles.

These patterns likely reflect:

- Coastal wind circulation changes
- Monsoon-related atmospheric conditions
- Seasonal temperature fluctuations
- Regional pollutant transport mechanisms

Importance of Pollution Spike Detection

The high ranking of the high_pollution_flag feature demonstrates the significance of sudden pollution events.

The model actively leverages anomaly detection indicators to identify abnormal environmental conditions before generating future forecasts.

Influence of Chemical Pollutants

Variables such as ozone, sulphur_dioxide, carbon_monoxide, and nitrogen_dioxide appear prominently among the top-ranked features.

This indicates that the forecasting model does not rely solely on particulate matter concentrations but instead learns complex interactions among multiple atmospheric pollutants.

Significance of Historical Trends

Lag variables, rolling statistics, moving averages, and volatility indicators collectively contribute substantial predictive power.

Their importance confirms that previous pollution behavior is one of the strongest indicators of future AQI conditions, reinforcing the suitability of the time-series forecasting approach adopted within the SkyCast platform.

Overall, the feature importance analysis validates the effectiveness of the engineered feature store and demonstrates that the model successfully captures both short-term environmental fluctuations and long-term seasonal pollution patterns characteristic of Karachi's urban atmosphere.

PART 4: EXPLAINABLE AI (XAI) THROUGH SHAP INTERPRETATION

4.1 Foundations of Model Interpretability

Although ensemble-based models such as the Extra Trees Regressor achieve high predictive accuracy, their internal decision-making process is inherently complex and difficult to interpret. In high-impact domains such as public health and environmental monitoring, it is essential that model predictions are transparent, explainable, and trustworthy.

To address this requirement, the SkyCast platform incorporates **SHAP (SHapley Additive exPlanations)** as its primary explainability framework.

SHAP provides a theoretically grounded method for interpreting machine learning models by computing **Shapley values**, which quantify the individual contribution of each feature to a specific prediction. This ensures that every AQI forecast can be decomposed into meaningful environmental drivers, allowing verification that the model is relying on physically consistent atmospheric relationships rather than spurious correlations.

4.2 SHAP Summary Global Interpretation (24-Hour Horizon)

The 24-hour forecasting horizon represents the most immediate prediction window and is primarily influenced by real-time pollutant concentrations, short-term environmental fluctuations, and seasonal indicators.

The SHAP global summary analysis can be interpreted as:

Feature Value (High = Red, Low = Blue) → Impact on Predicted AQI

Primary Particulate Matter Indicators

The most dominant contributors in the 24-hour horizon are fine particulate matter features such as:

- pm2_5
- pm25_log

High values of these features consistently correspond to strong positive SHAP values, indicating a direct and proportional increase in predicted AQI levels. This confirms that elevated particulate concentrations are the strongest drivers of immediate air quality deterioration.

Conversely, lower pollutant values generate negative SHAP contributions, shifting predictions toward cleaner air classifications.

Pollution Volatility Indicators

The **high_pollution_flag** feature plays a significant role in detecting abrupt environmental changes. When activated, it produces a strong positive SHAP impact, enabling the model to respond effectively to sudden pollution spikes and hazardous atmospheric events.

Seasonal and Climatic Effects

Cyclical variables such as:

- month_cos
- season

demonstrate that the model successfully captures Karachi's seasonal air quality behavior. Certain seasonal conditions, particularly winter-related atmospheric inversions, reduce pollutant dispersion and lead to higher predicted AQI values.

4.3 Middle-Horizon Shift Analysis (48-Hour Horizon)

As the forecasting horizon extends to 48 hours, the model gradually shifts its reliance from real-time measurements to historical trends and chemical persistence patterns.

Persistent Air Quality Trends

Key lag-based features such as:

- aqi_lag_24h
- pm25_lag_24

become significantly more influential. These variables indicate that air quality conditions exhibit strong temporal persistence, where current atmospheric states are heavily dependent on previous-day pollution levels.

Higher values of these lag features correspond to increased AQI predictions, reinforcing the continuity of pollution patterns across consecutive days.

Role of Secondary Pollutants

Trace gases including:

- ozone
- sulphur_dioxide

gain importance in the 48-hour horizon. Elevated concentrations of these pollutants act as early indicators of sustained smog formation and atmospheric instability.

Baseline Pollution Constraints

The feature **pm25_roll_min_24** serves as a lower-bound indicator of pollution levels. Even when short-term improvements occur, a high rolling minimum suggests persistent background pollution, preventing the model from underestimating future AQI conditions.

4.4 Long-Horizon Stability Analysis (72-Hour Horizon)

At the 72-hour forecasting horizon, the model transitions from short-term variability to long-term atmospheric and meteorological structures. Predictions at this stage are primarily governed by macro-scale environmental conditions.

Atmospheric Persistence and Long-Term Trends

Features such as:

- pm25_lag_48
- pm25_roll_max_24

become dominant predictors. These variables help the model determine whether the region is experiencing sustained pollution accumulation or gradual atmospheric recovery.

Meteorological Stability Factors

Long-term weather patterns gain increased importance, particularly:

- temperature_roll_mean_24
- humidity_roll_mean_24

Higher values of these variables are often associated with improved dispersion conditions, leading to negative SHAP contributions and reduced predicted AQI values.

Wind and Pressure Dynamics

Key atmospheric circulation variables include:

- wind_speed_roll_mean_24
- pressure_roll_mean_24

Stronger wind speeds consistently produce negative SHAP effects, reflecting improved pollutant dispersion due to coastal air movement. In contrast, elevated barometric pressure contributes positively to SHAP values, indicating atmospheric stagnation that traps pollutants near the surface.

4.5 Multi-Horizon Interpretability Summary

The SHAP analysis reveals a clear and physically consistent evolution in model behavior across forecasting horizons:

- **24-Hour Horizon:** Dominated by real-time pollutant concentrations and immediate atmospheric conditions.
- **48-Hour Horizon:** Balanced influence between lagged pollution trends and secondary chemical indicators.
- **72-Hour Horizon:** Governed primarily by long-term meteorological patterns and atmospheric stability factors.

This progression demonstrates that the Extra Trees model not only achieves high predictive accuracy but also aligns closely with known atmospheric science principles governing urban air pollution dynamics in coastal environments such as Karachi.

By integrating SHAP-based interpretability, SkyCast ensures transparency, accountability, and trustworthiness in its AQI forecasting outputs, making it suitable for public health decision support systems.

PART 5: LOCAL INTERPRETABILITY VIA LIME EXPLANATIONS

5.1 Objectives of Localized Instance Explanations

While SHAP provides a global perspective on feature importance across the entire 3.25-year dataset, real-world decision-making in environmental monitoring requires instance-level interpretability. In practical scenarios, stakeholders such as policymakers, healthcare professionals, and citizens need to understand why a specific AQI prediction was generated at a particular time.

To address this requirement, the SkyCast platform integrates **LIME (Local Interpretable Model-Agnostic Explanations)** for local prediction interpretation.

LIME operates by constructing a simplified, interpretable linear model around a single prediction instance. This localized surrogate model approximates the behavior of the complex machine learning model in the vicinity of the selected data point. As a result, it provides a transparent explanation of the immediate environmental factors influencing a specific AQI forecast.

5.2 Decoupling the Dashboard Visual Matrix

Within the Streamlit dashboard interface, LIME outputs are presented as intuitive, color-coded horizontal contribution bars. These visual explanations allow users to quickly interpret the direction and magnitude of each feature's influence on the predicted AQI value.

Interpretation of LIME Contribution Bars

- **Orange Contribution Bars (Weight > 0):**
- These represent features that increase the predicted AQI value. They indicate environmental conditions contributing to worsening air quality, such as elevated pollution levels or unfavorable atmospheric conditions.

- **Blue Contribution Bars (Weight ≤ 0):**

These represent features that decrease the predicted AQI value. They highlight environmental factors that improve air quality or promote pollutant dispersion.

This visualization framework ensures that non-technical users can interpret complex model behavior without requiring advanced statistical knowledge.

5.3 Analysis of Primary Risk Ingestion (Case Study 1)

The first LIME-based instance analysis represents a scenario where the model predicts a high AQI level, indicating poor air quality conditions.

Sustained Baseline Pollution

The feature **high_pollution_flag (> 0.00)** produces the most dominant positive contribution in this instance. Its strong influence indicates the presence of an ongoing regional pollution event, which significantly elevates the predicted AQI level.

Chemical Saturation Effects

Key particulate matter features such as:

- **pm25_log (> 3.44)**
- **pm2_5 (> 31.10)**

generate substantial positive contributions. These elevated values confirm that fine particulate concentration is a primary driver of poor air quality in this scenario.

Secondary Pollutant Contributions

Additional chemical pollutants including:

- **ozone (> 31.84)**
- **sulphur_dioxide (> 13.91)**

provide smaller yet meaningful positive contributions. These features reflect industrial emissions and secondary chemical reactions that intensify atmospheric pollution levels.

Collectively, these factors push the local prediction toward a higher-risk AQI category.

5.4 Analysis of Atmospheric Mitigation (Case Study 2)

The second LIME-based instance represents a transitional atmospheric state where environmental conditions contribute to improved air quality.

Seasonal and Cyclical Influence

The feature **month_cos (> 0.50)** continues to exert a positive baseline influence on AQI levels, reflecting seasonal conditions that may still favor pollutant accumulation in certain periods.

Reduction in Particulate Concentration

In contrast to the previous case, the following features show reduced values:

- `pm25_log` (≤ 3.44)
- `pm2_5` (≤ 31.10)

These variables generate strong negative contributions, indicating that lower particulate matter concentrations are the most significant drivers of improved air quality in this scenario.

Meteorological Dispersion Effects

Additional features such as:

- `ozone` (≤ 31.84)
- `temperature_roll_mean_24` (≤ 25.43)

produce negative contributions to the prediction. These conditions reflect atmospheric processes such as coastal airflow and temperature stabilization, which enhance pollutant dispersion and reduce overall AQI levels.

5.5 Combined Interpretability Framework

By integrating SHAP-based global interpretability with LIME-based local explanations, the SkyCast platform achieves a dual-layer transparency framework:

- **SHAP (Global Interpretability):** Explains overall model behavior across the full dataset, identifying long-term environmental drivers.
- **LIME (Local Interpretability):** Explains individual predictions in real time, enabling instance-level decision transparency.

This combined approach ensures that the forecasting system is not only accurate but also interpretable, trustworthy, and suitable for deployment in public health and environmental decision-support applications.

As a result, SkyCast moves beyond black-box machine learning systems and provides a transparent, evidence-based AQI forecasting platform capable of supporting informed decision-making at both macro and micro temporal scales.

PART 6: PRODUCTION USER INTERFACE & COMPREHENSIVE DASHBOARD ECOSYSTEM

6.1 Architectural Overview of the User Interface

The user-facing layer of the SkyCast platform is implemented as an interactive web application using **Streamlit**, located in `dashboards/streamlit_app.py`. This interface serves as the primary medium through which predictive insights are delivered to end users.

The dashboard is designed using a custom visual theme with clean pastel styling to ensure clarity, readability, and ease of interpretation. Rather than relying on generic templates, the interface is structured to support high-density scientific outputs in a visually intuitive manner.

This design approach bridges the gap between complex machine learning computations and real-world usability, enabling both technical and non-technical users to effectively interpret air quality trends, forecasts, and health risk indicators.

6.2 Real-Time Hazard Warnings and Categorization Mechanics

To improve accessibility for general users, the platform includes an automated **Public Health Hazard Alert Engine**. This module translates raw AQI predictions into standardized health categories aligned with regulatory air quality thresholds.

Each predicted value is evaluated against predefined AQI breakpoints, dynamically updating the dashboard's visual components, alert banners, and recommended safety actions.

Table 6.1: Air Quality Index (AQI) Categorization and Health Advisory System

AQI Range	Category	Visual Indicator	Public Advisory	Health
0 – 50	Good	Cool Sky Blue	Air quality is satisfactory with minimal or no health risk.	
51 – 100	Moderate	Pastel Emerald	Air quality is acceptable; sensitive individuals should limit prolonged outdoor exposure.	
101 – 150	Unhealthy for Sensitive Groups	Warm Amber	Sensitive individuals may experience symptoms; general population remains mostly unaffected.	
151 – 200	Unhealthy	Deep Magenta	Increased health risks for the general population; reduce prolonged outdoor	

			exposure.
201+	Very Unhealthy	Dark Purple	Hazardous conditions; outdoor exposure should be minimized and air filtration is recommended.

This structured classification system ensures that users can instantly understand the severity of air quality conditions without requiring technical interpretation of numerical AQI values.

6.3 Interactive Index Charting Matrix

To enhance temporal understanding of air quality trends, the dashboard incorporates a multi-horizon interactive visualization system. This charting matrix allows users to observe both historical patterns and future predictions simultaneously.

Key Features of the Visualization System

Multi-Horizon Forecast Display

The timeline chart integrates:

- Real-time observed AQI values
- 24-hour forecast predictions
- 48-hour forecast predictions
- 72-hour forecast predictions

This enables continuous comparison between current conditions and projected environmental trends.

Regulatory Threshold Markers

Horizontal reference lines corresponding to AQI categories are embedded within the visualization. These markers allow users to quickly identify when predicted values cross into elevated risk zones.

Interactive Data Exploration

The chart supports dynamic hover-based tooltips, enabling users to inspect:

- Exact AQI values
- Timestamp-specific predictions
- Underlying meteorological conditions

This interactive design enhances interpretability and improves user engagement with environmental data.

6.4 Analytical Intelligence and Export Pipelines

To support advanced analysis, regulatory reporting, and research applications, SkyCast includes integrated data export and reporting utilities within the dashboard sidebar.

System Functions

- **Generate AI Forecast → Executive Report Generation**
- **Export Prediction Ledger → Time-Series Data Download (CSV)**

These tools enable seamless transition from real-time forecasting to offline analysis and institutional reporting.

6.4.1 Automated Executive Report Generator ("Generate AI Forecast")

When activated, the **Generate AI Forecast** feature initiates an automated reporting pipeline that synthesizes live environmental data with machine learning predictions.

This process retrieves data from Supabase storage layers and combines it with model outputs to generate a structured executive-level report. The generated document includes:

- Multi-day AQI forecasts
- Health risk summaries
- Trend analysis interpretations
- Automated environmental warnings

The output is formatted for institutional use, making it suitable for policymakers, environmental agencies, and public health organizations.

6.4.2 Flattened Time-Series Data Export ("Save as CSV")

For researchers and data analysts, the platform provides a direct export mechanism that downloads the full environmental dataset in CSV format. This dataset includes:

- Raw meteorological variables
- Pollutant concentration measurements
- Engineered machine learning features
- Multi-horizon AQI predictions

Inverse Transformation Mechanism

During export generation, the system applies a reverse transformation function (**backcalculate_pm25**) to reconstruct continuous pollutant concentration values from predicted AQI outputs. This ensures that exported datasets retain both predictive and physically interpretable components.

As a result, the exported file serves as a complete analytical ledger that supports further research, validation studies, and secondary model development.

6.5 Final System Summary

The SkyCast dashboard ecosystem successfully transforms complex machine learning predictions into a fully interactive, user-friendly environmental intelligence platform. Through the integration of real-time forecasting, explainable AI, interactive visualization, and automated reporting tools, the system provides a comprehensive solution for air quality monitoring and decision support.

By combining scientific rigor with accessible design, the platform ensures that critical environmental insights are available not only to technical users but also to the general public, thereby enhancing awareness, preparedness, and public health responsiveness in urban environments.